7457 - Quenching physics and age demographics of stellar populations in a massive radio-loud AGN host galaxy at z~3.5

Cycle: 4, Proposal Category: GO

INVESTIGATORS

Name	*Institution*
Dr. Wuji Wang (PI)	**California Institute of Technology**
Dr. Dominika Wylezalek (CoI) (ESA Member) (CoPI)	Universitat Heidelberg
Dr. Joel Vernet (CoI) (ESA Member)	European Southern Observatory - Germany
Dr. Carlos Albert De Breuck (CoI) (ESA Member)	European Southern Observatory - Germany
Dr. Nadia L Zakamska (CoI)	The Johns Hopkins University
Prof. David Rupke (CoI)	Rhodes College
Dr. Matthew D. Lehnert (CoI) (ESA Member)	Centre de Recherche Astrophysique de Lyon
Dr. Nicole Nesvadba (CoI) (ESA Member)	Observatoire de la Cote d'Azur
Dr. Andrey Vayner (CoI)	California Institute of Technology
Dr. Daniel K Stern (CoI)	Jet Propulsion Laboratory
Dr. Lee Armus (CoI) (US Admin CoI)	California Institute of Technology

OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
Observation Folder				
	1	MIRI_MRS_TNJ0121	MIRI Medium Resolution Spectroscopy	(1) TNJ0121+1320
	2	MIRI_MRS_TNJ0121_BKG	MIRI Medium Resolution Spectroscopy	(2) TNJ0121+1320-BKG
	3	NIRCam_TNJ0121	NIRCam Imaging	(3) TNJ0121+1320-NIRCam

ABSTRACT

Feedback from actively accreting supermassive black holes (AGN) impact their host galaxies in multiple ways which can be traced on pc to kpc scales. Quenching is frequently modeled as heating up the gas such that it can no longer form stars.

We propose a pilot MIRI/MRS observation targeting the warm molecular gas (H2) in a high-redshift radio galaxy (J0121) at z=3.5. We also propose NIRCam imaging to cover the old stars in the host and scattered blue AGN light.

J0121 already has robust evidence for quasar-mode and radio-mode feedback and merger process through the study of optical, UV and submm emission lines. JWST/MIRI will open the study of a crucial gas phase of active galaxies at the onset of Cosmic Noon to address the long-standing question on how and when AGN quench massive galaxies. The 3D coverage with MIRI MRS will provide a unique opportunity to map the warm H2, a previously inaccessible ISM tracer in HzRGs at z>3. The diffraction-limited resolution of MIRI and NIRCam photometry will cover the broad spectral energy distribution from restframe 0.1 to 6μm, determining the spatial distribution of old stellar population on sub-kpc scales, isolate the AGN-heated dust emission, and constrain the young stars beneath AGN.

This proposal will tell us if the AGN can prevent warm H2 from cooling efficiently and determine the age (and spatial distribution) of all stars. J0121 has a plethora of multi-wavelength data (incl. JWST NIRSpec IFU for the warm ionized gas and ALMA for cold gas) and is ideal for linking the energy output of AGN through different phases and studying its impact on the ingredients of the host galaxies—the warm H2 and stars—of powerful AGN at z=3.5.

OBSERVING DESCRIPTION

We will observe one high-redshift radio galaxies at z=3.519 using the MIRI MRS in MEDIUM sub-band. The main goal is to spatially map the warm H_2 tracers, several will be captured in CH2 with MEDIUM sub-band, including the brightest, H_2 1-0 S(1). The PAH3.3 feature will also be captured in CH3 MEDIUM.

We will also setup NIRCam imaging at F250M, F335M, F410M, and F480M to sample the stellar SED. The F070W, F090W, F115W, and F140M at the shorter wavelength will be selected simulatenously to cover the blue AGN light in rest-frame UV.

The following observing modes are:

TNJ0121+1320 -- MIRI MRS, MEDIUM sub-band:

4-point dither pattern

25 groups and 7 integration per dither position

Readout mode: SLOWR1

--> total exposure time on source: 4.8 hours

1 background image slightly (~1 arcmin) offset from the target position with readout 12 groups (<300s minimizing CR) and 14 integrations, science exposure: 4.8 hours (within the MRS exposure)

Background limited to have <30 percentile

TNJ0121+1320 -- NIRCam imaginge:

4 filters combinations:

F070W+F250M

F090W+F335M

F115W+F410M

F140M+F480M

For each, INTRASCA 4 SMALL STANDARD dither pattern, 5 groups and 1 integration per dither position

-->total on source: 1.0 hours

TNJ0121+1320 -- MIRI MRS background (source position will be fine tuned):

4-point dither pattern

same setup as at per science dither position

--> total on source time: 4.8 hours

1 background image with F2550W filter to cover the science target to facilitate SED fitting: 12 groups and 14 inegrations, science exposure: 4.8 hours (within MRS exposure)

PA constrained to cover the science target

Total requested observing time: 5.8 hours science time, 4.2 hours overhead, 4.8 hours total background exposure

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	TNJ0121+1320	RA: 01 21 42.7296 (20.4280400d) Dec: +13 20 58.09 (13.34947d) Equinox: J2000		
Comments: This object was generated by the targetselector and retrieved from the NED database. Exact position may be fine-tuned due to available schedule (PA) Category=Galaxy Description=[Active galactic nuclei, Active galaxies, High-redshift galaxies, Radio galaxies, Ultraluminous infrared galaxies] Extended=YES				
(2)	TNJ0121+1320-BKG	RA: 01 21 42.3400 (20.4264167d) Dec: +13 21 56.29 (13.36564d) Equinox: J2000		
Comments: Background of TNJ0121+1320 Exact position may be fine-tuned due to available schedule (PA) Update September 3rd 2025, update background position and constrain background PA Category=Calibration Description=[Telescope/sky background]				
(3)	TNJ0121+1320-NIRCam	RA: 01 21 42.7300 (20.4280417d) Dec: +13 20 58.09 (13.34947d) Equinox: J2000		
Comments: Category=Galaxy Description=[Active galaxies, Elliptical galaxies, Emission line galaxies, High-redshift galaxies, Radio loud quasars]				

Observation

Proposal 7457, Observation 1: MIRI_MRS_TNJ0121

Wed Oct 08 13:00:09 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observations:[MIRI_MRS_TNJ0121_BKG (Obs 2)]

Diagnostics

(MIRI_MRS_TNJ0121 (Obs 1)) Warning (Form): Filter mismatch between science and background observations may result in incorrect background subtraction.

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	TNJ0121+1320	RA: 01 21 42.7296 (20.4280400d) Dec: +13 20 58.09 (13.34947d) Equinox: J2000		

Comments: This object was generated by the targetselector and retrieved from the NED database.
Exact position may be fine-tuned due to available schedule (PA)
Category=Galaxy
Description=[Active galactic nuclei, Active galaxies, High-redshift galaxies, Radio galaxies, Ultraluminous infrared galaxies]
Extended=YES

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
F1000W	All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	EXTENDED SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F770W	SLOWR1	12	14	1	Dither 1	4	56	17296.302	
1	MEDIUM(B)	MRSLONG		SLOWR1	25	7	1	Dither 1	4	28	17296.302	265307
1	MEDIUM(B)	MRSSHORT		SLOWR1	25	7	1	Dither 1	4	28	17296.302	265307

Special Requirements

Background Limited. Background no more than 30th percentile above minimum

Sequence Observations 1, 2, Non-interruptible

Observation

Proposal 7457, Observation 2: MIRI_MRS_TNJ0121_BKG

Wed Oct 08 13:00:09 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observation For: [MIRI_MRS_TNJ0121 (Obs 1)]

Diagnostics

(MIRI_MRS_TNJ0121_BKG (Obs 2)) Warning (Form): Imager Filter overlap.

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	TNJ0121+1320-BKG	RA: 01 21 42.3400 (20.4264167d) Dec: +13 21 56.29 (13.36564d) Equinox: J2000		

Comments: Background of TNJ0121+1320
Exact position may be fine-tuned due to available schedule (PA)
Update September 3rd 2025, update background position and constrain background PA
Category=Calibration
Description=[Telescope/sky background]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	BACKGROUND	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F2550W	SLOWR1	12	14	1	Dither 1	4	56	17296.302	265307
1	MEDIUM(B)	MRSLONG		SLOWR1	25	7	1	Dither 1	4	28	17296.302	
1	MEDIUM(B)	MRSSHORT		SLOWR1	25	7	1	Dither 1	4	28	17296.302	

Special Requirements	
	Aperture PA Range 22.78 to 75.6 Degrees (V3 22.78 to 75.6) Sequence Observations 1, 2, Non-interruptible

Observation

Proposal 7457, Observation 3: NIRCam_TNJ0121

Wed Oct 08 13:00:09 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Diagnostics

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	TNJ0121+1320-NIRCam	RA: 01 21 42.7300 (20.4280417d) Dec: +13 20 58.09 (13.34947d) Equinox: J2000		

Comments:
Category=Galaxy
Description=[Active galaxies, Elliptical galaxies, Emission line galaxies, High-redshift galaxies, Radio loud quasars]

Template

Module	Subarray	Target Placement
ALL	FULL	Module B (B4 corner)

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	INTRASCA	4	STANDARD	8" (SMALL)	1

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
1	F070W	F250M	SHALLOW4	5	1	4	4	1030.73	265307
2	F090W	F335M	SHALLOW4	5	1	4	4	1030.73	265307
3	F115W	F410M	SHALLOW4	5	1	4	4	1030.73	265307
4	F140M	F480M	SHALLOW4	5	1	4	4	1030.73	265307

Special Requirements

Aperture PA Range 0.05262691 to 11.05262691 Degrees (V3 0.0 to 11.0)
Aperture PA Range 13.05262691 to 226.05262691 Degrees (V3 13.0 to 226.0)
Aperture PA Range 230.05262691 to 359.05262691 Degrees (V3 230.0 to 359.0)
Offset 28.634675660810597 arcsec, 49.607102393847185 arcsec
Fiducial Point Override NRCBS_FULL